
Hamiltonian Asymmetric Fusion: One-Way Safe Directed Refinement under Modality Imbalance

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Abstract

In RGB–D salient object detection, multimodal fusion is commonly implemented via symmetric token interaction, implicitly allowing information to flow in both directions. Under *modality imbalance*—when an auxiliary stream is substantially noisier than a designated primary stream—such symmetry permits a *backflow channel* through which auxiliary noise can affect the primary representation and accumulate across iterative refinement stages. We formulate fusion in this regime as *directed refinement with one-way safety*: the primary modality defines a guidance field, while the auxiliary representation is iteratively refined and auxiliary-induced primary perturbations are explicitly quantified. We propose *Hamiltonian Asymmetric Fusion* (HAF), a lightweight unrolled refinement block that updates auxiliary tokens through Hamiltonian-inspired, momentum-regularized dynamics with gated driving. The refinement force is instantiated by FFT-based spectral global correlation and modulated by a shared learnable spectral response to emphasize reliable frequency components with minimal parameters; a leaky momentum gate damps stale updates during multi-step refinement. Experiments on six RGB–D SOD benchmarks show consistent gains and substantially more graceful degradation under controlled auxiliary corruption.

1. Introduction

Multimodal learning improves robustness by integrating complementary cues across heterogeneous sensors (Baltrušaitis et al., 2018). In RGB–D salient object detection (SOD), depth provides geometric structure that helps delineate object boundaries and resolve textureless regions, while RGB supplies appearance cues for semantic discrimination (Zhou et al., 2021a; Chen et al., 2025). Modern RGB–D SOD systems typically rely on progressive decoding and stage-wise refinement, where cross-modal interaction is repeatedly applied across feature hierarchies (Sun et al., 2023; Hu et al., 2024; Luo et al., 2024).

Modality imbalance induces backflow under iterative fusion. While prior work often emphasizes modality gaps and alignment, we focus on a different failure mode that becomes critical under repeated refinement: *modality imbalance*, where an auxiliary stream is substantially noisier or less reliable than a designated primary stream. In this regime, *symmetric* fusion (e.g., bidirectional cross-attention or symmetric diffusion-style coupling) permits a *backflow channel*: auxiliary noise can enter the primary representation and accumulate across refinement steps, potentially leading to unstable refinement trajectories and degraded dense predictions (Wang & Yu, 2025). Thus, the central problem is to design fusion operators that remain stable across multi-step refinement in the presence of highly asymmetric modality reliability.

Directed refinement with one-way safety. We argue that fusion under modality imbalance should be formulated as a *directed refinement process* with an explicit safety requirement. The primary modality should define a guidance field for refinement, while the auxiliary representation is iteratively refined; meanwhile, auxiliary-induced changes to the primary stream should be structurally excluded or explicitly bounded. We refer to exact independence of the primary output from the auxiliary input as *one-way safety*; when a nonzero auxiliary-to-primary residual is retained, we instead quantify the resulting primary perturbation. This contrasts with symmetric interaction operators—including cross-attention and Cross-Diffusion Attention (CDA) (Wang et al., 2024)—which structurally permit backflow and do not explicitly encode role-asymmetric influence across multiple

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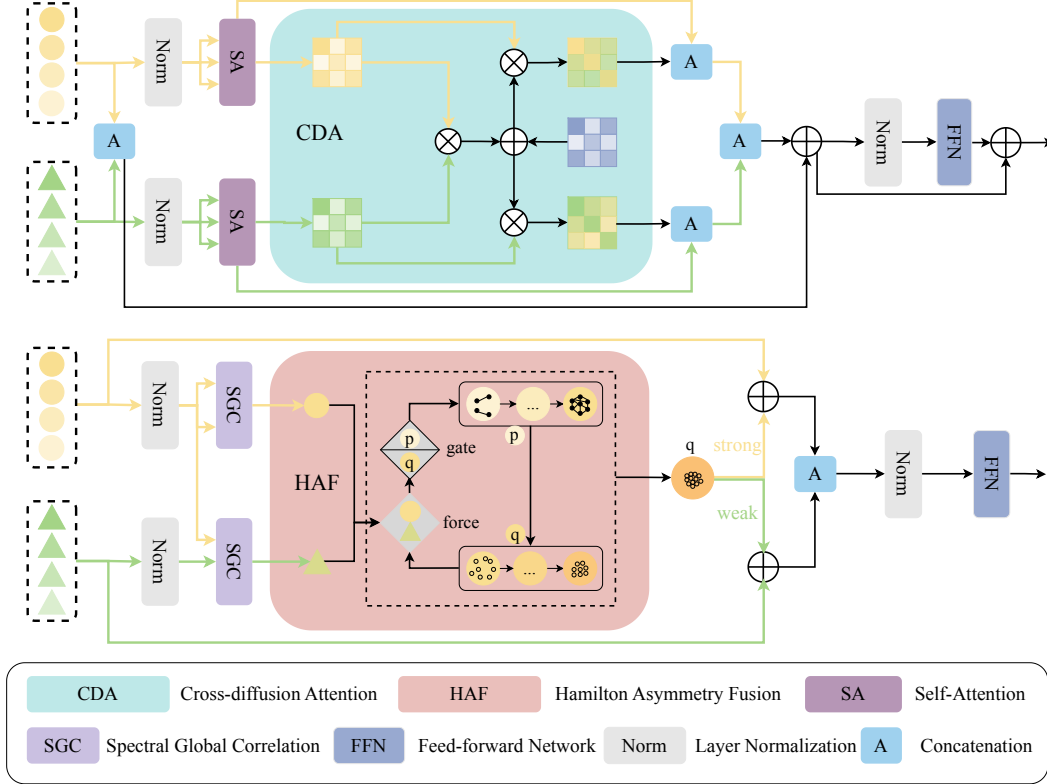


Figure 1. Comparison between CDA and HAF. **Top:** CDA performs *symmetric* cross-diffusion by mutually propagating information between modality-specific attention maps. **Bottom:** HAF formulates the fusion process as an unrolled Hamiltonian-inspired momentum refinement with *asymmetric* residual connections.

refinement steps. Fig. 1 provides a schematic comparison of symmetric interaction versus role-asymmetric refinement.

Hamiltonian Asymmetric Fusion (HAF). Based on this principle, we propose **Hamiltonian Asymmetric Fusion (HAF)**, a lightweight fusion block that implements directed refinement via an unrolled, Hamiltonian-inspired momentum update on auxiliary tokens driven by a primary-guided force field. HAF combines (i) a global correlation-based driving force with spectral adaptation and (ii) leaky gated momentum to regulate multi-step refinement. Importantly, HAF is *role-asymmetric*: the primary stream acts as an anchor that induces refinement, while only the auxiliary state is iteratively updated; the refined auxiliary information is then injected back with asymmetric residual weights to reduce cross-modal interference. In our RGB-D SOD setting, we use a task-defined role assignment (depth as primary, RGB as auxiliary), while the formulation can be applied under alternative role assignments.

Results. We evaluate HAF on six RGB-D SOD benchmarks and observe consistent improvements in S -measure, $\max F_\beta$, $\max E_\xi$, and MAE over strong Transformer-based baselines. We further conduct unified module-swapping and

controlled auxiliary-corruption experiments to isolate fusion behavior under modality imbalance, where HAF demonstrates more graceful degradation than competing fusion plugins.

Contributions.

- We identify *modality imbalance* under iterative refinement as a key failure mode and propose *directed refinement with one-way safety* to structurally exclude auxiliary-to-primary backflow when the corresponding residual is disabled and to quantify it otherwise.
- We introduce *HAF*, a role-asymmetric fusion block that updates only auxiliary tokens through unrolled momentum-regularized refinement driven by the primary modality.
- We validate consistent gains and improved robustness via unified module-swapping and controlled auxiliary corruption across six benchmarks.

Conflict of Interest Disclosure. The authors declare no financial conflicts of interest related to this work.

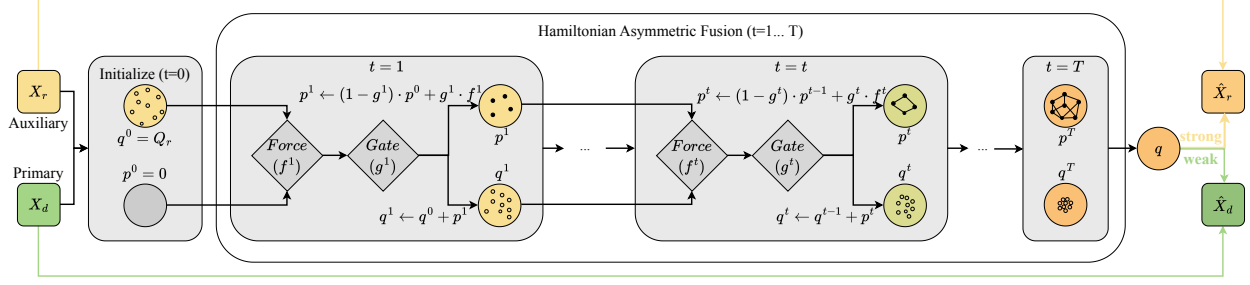


Figure 2. Overall pipeline of HAF. Starting from $\mathbf{q}^0 = \mathbf{Q}_r$ and $\mathbf{p}^0 = \mathbf{0}$, HAF unrolls T steps where a force term is computed via spectral global correlation and modulated by a learned gate to update $(\mathbf{p}^t, \mathbf{q}^t)$. The final state $\mathbf{q}^T$ is injected back to both modalities with asymmetric residual weights.

2. Method

We study feature fusion under *unequal reliability* between two modalities, e.g., RGB as a noisy *auxiliary* stream and depth as a cleaner *primary* stream. Let $\mathbf{X}_r, \mathbf{X}_d \in \mathbb{R}^{n \times d}$ denote token maps flattened over space, where $n = HW$ and d is the feature dimension. Most existing fusion blocks rely on *symmetric* bidirectional interactions (e.g., cross-attention), implicitly assuming comparable signal-to-noise ratio (SNR) across modalities. Under *modality imbalance*, where $\mathbf{X}_r$ is substantially noisier, symmetric coupling can trade auxiliary utility against *noise backflow* into the cleaner primary stream $\mathbf{X}_d$, thereby degrading primary representations.

We introduce *Hamiltonian Asymmetric Fusion* (HAF), a directed interaction mechanism that separates primary-guided refinement from the optional auxiliary-to-primary residual path. HAF treats the primary stream as a fixed *guidance field* and iteratively refines only the auxiliary state within this field. Crucially, when the optional auxiliary-to-primary residual is disabled, the primary output is exactly independent of auxiliary features; when it is retained, its influence is quantified explicitly in Corollary 3.6. Fig. 2 presents the overall architecture.

2.1. Spectral Global Correlation

To capture long-range structure without instantiating a dense $n \times n$ affinity matrix, the design follows GFNet’s Fourier-domain global mixing paradigm (Rao et al., 2021). The input-independent global filter is instantiated as a query-key spectral correlation operator (denoted as *SGC* in this work), and a *shared learnable spectral response* is incorporated to reweight frequency components, prioritizing reliable bands under auxiliary corruption.

Given tokens $\mathbf{X} \in \mathbb{R}^{n \times d}$, linear projections yield $\mathbf{Q}, \mathbf{K} \in \mathbb{R}^{n \times d}$, which are reshaped into $H \times W$ maps per channel and head. Let $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the 2D FFT and its inverse, applied independently to each channel (and head). Throughout the analysis in Section 3 and Appendix C, $\mathcal{F}$ is

taken to be unitary. Under any other fixed FFT normalization, the same results hold after including the corresponding deterministic scale factors in the displayed operator bounds. A Hann window $\mathbf{w} \in \mathbb{R}^{H \times W}$ is applied to reduce boundary artifacts and spectral leakage. We further introduce a learnable frequency response $\mathbf{G} \in \mathbb{R}^{H \times W}$ that is *shared across channels and heads*, adding only $O(HW)$ parameters while enabling learnable frequency-band reweighting.

We define the real-valued (augmented) SGC operator as

$$\text{SGC}(\mathbf{Q}, \mathbf{K}) = \text{Re} \left\{ \mathcal{F}^{-1} \left(\mathbf{G} \odot \mathcal{F}(\mathbf{Q} \odot \mathbf{w}) \odot \overline{\mathcal{F}(\mathbf{K} \odot \mathbf{w})} \right) \right\}. \quad (1)$$

where $\odot$ denotes element-wise multiplication and $\overline{(\cdot)}$ is complex conjugation. The real-part operator makes the real-valued feature convention explicit; it is redundant whenever the learned frequency response is parameterized with the Hermitian symmetry required by a real inverse FFT. Eq. (1) computes dense correlation fields in $O(n \log n)$ time and $O(n)$ memory, avoiding explicit construction of $n \times n$ attention weights.

We use SGC both intra-modally and cross-modally. Within-modality correlations are $\mathbf{U}_r = \text{SGC}(\mathbf{Q}_r, \mathbf{K}_r)$ and $\mathbf{U}_d = \text{SGC}(\mathbf{Q}_d, \mathbf{K}_d)$. For directed cross-modal interaction, we compute depth-to-RGB correlation

$$\mathbf{U}_{r \leftarrow d} = \text{SGC}(\mathbf{Q}_r, \mathbf{K}_d), \quad (2)$$

which correlates auxiliary queries with primary keys and will be used to instantiate the primary-induced force field for one-way refinement.

2.2. Hamiltonian Asymmetric Fusion

We reinterpret fusion as a controlled refinement process in which *only* the auxiliary tokens evolve. The Hamiltonian construction below serves as a design motivation for introducing position-momentum variables; the implemented learned force is not assumed to be an exact conservative gradient, and our analysis does not rely on energy conservation

or exact symplecticity. Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ denote the auxiliary state to be refined and $\mathbf{p} \in \mathbb{R}^{n \times d}$ its momentum. We define a Hamiltonian

$$H(\mathbf{q}, \mathbf{p}) = \frac{1}{2} \|\mathbf{p}\|^2 + V(\mathbf{q}), \quad (3)$$

where $V(\mathbf{q})$ denotes an idealized potential energy induced by correlations with the primary modality. The corresponding continuous-time Hamiltonian dynamics are

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}} = \mathbf{p}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}} = -\nabla_{\mathbf{q}} V(\mathbf{q}). \quad (4)$$

Spectral force field. We instantiate the force using SGC. Let $\mathbf{K}_d$ and $\mathbf{K}_r$ denote key projections from the primary and auxiliary streams. We define

$$F_{\text{main}}(\mathbf{q}) = \text{SGC}(\mathbf{q}, \mathbf{K}_d), \quad (5)$$

$$F_{\text{aux}}(\mathbf{q}) = \text{SGC}(\mathbf{q}, \mathbf{K}_r), \quad (6)$$

Motivated by the negative potential-gradient term in the idealized dynamics, we instantiate the implemented force as

$$\mathbf{f}(\mathbf{q}) = F_{\text{main}}(\mathbf{q}) - \mathbf{q} + \gamma(\mathbf{q} - F_{\text{aux}}(\mathbf{q})), \quad (7)$$

where $\gamma \geq 0$ balances (i) attraction toward primary-supported patterns through $F_{\text{main}}(\mathbf{q}) - \mathbf{q}$ and (ii) repulsion from auxiliary-only patterns through $\mathbf{q} - F_{\text{aux}}(\mathbf{q})$. Intuitively, the second term suppresses structures that are self-consistent within the auxiliary stream but weakly supported by the primary.

Damped refinement with leaky momentum gating. We use a semi-implicit position update together with a learned gate that regulates multi-step refinement. We initialize $\mathbf{q}^0 = \mathbf{Q}_r$ and $\mathbf{p}^0 = \mathbf{0}$. For $t = 1, \dots, T$, define the force

$$\mathbf{f}^t = F_{\text{main}}(\mathbf{q}^{t-1}) - \mathbf{q}^{t-1} + \gamma(\mathbf{q}^{t-1} - F_{\text{aux}}(\mathbf{q}^{t-1})), \quad (8)$$

and compute a gate

$$\mathbf{g}^t = \sigma(\text{MLP}(\mathbf{q}^{t-1} \parallel \mathbf{p}^{t-1})), \quad \mathbf{g}^t \in (0, 1)^{n \times d}, \quad (9)$$

where $\parallel$ denotes concatenation and $\sigma(\cdot)$ is the sigmoid.

Crucially, we adopt a *leaky* (gated) momentum update

$$\mathbf{p}^t = (\mathbf{1} - \mathbf{g}^t) \odot \mathbf{p}^{t-1} + \mathbf{g}^t \odot \mathbf{f}^t, \quad (10)$$

$$\mathbf{q}^t = \mathbf{q}^{t-1} + \mathbf{p}^t. \quad (11)$$

Compared to the pure accumulation $\mathbf{p}^t = \mathbf{p}^{t-1} + \mathbf{g}^t \odot \mathbf{f}^t$, the leaky form in (10) explicitly damps the contribution of stale momentum. The resulting update is represented exactly by the state map analyzed in Section 3; it is a leaky momentum refinement rather than an exact symplectic integrator. We use a small fixed unroll depth, with $T = 4$ in the reported experiments.

Algorithm 1 Hamiltonian Asymmetric Fusion (HAF) Block

Input: auxiliary features $\mathbf{X}_r \in \mathbb{R}^{n \times d}$, primary features $\mathbf{X}_d \in \mathbb{R}^{n \times d}$, steps T , repulsion weight γ , fusion weights $\alpha_{\text{aux}}, \alpha_{\text{main}}$

Output: fused features $\hat{\mathbf{X}}_r, \hat{\mathbf{X}}_d$

// **Projection and initialization**

$\mathbf{Q}_r, \mathbf{K}_r \leftarrow \text{LinearProj}(\mathbf{X}_r)$

$\mathbf{Q}_d, \mathbf{K}_d \leftarrow \text{LinearProj}(\mathbf{X}_d)$

$\mathbf{q}^0 \leftarrow \mathbf{Q}_r$

$\mathbf{p}^0 \leftarrow \mathbf{0}$

for $t = 1$ **to** T **do**

$F_{\text{main}} \leftarrow \text{SGC}(\mathbf{q}^{t-1}, \mathbf{K}_d)$

$F_{\text{aux}} \leftarrow \text{SGC}(\mathbf{q}^{t-1}, \mathbf{K}_r)$

$\mathbf{f}^t \leftarrow F_{\text{main}} - \mathbf{q}^{t-1} + \gamma(\mathbf{q}^{t-1} - F_{\text{aux}})$

$\mathbf{g}^t \leftarrow \sigma(\text{MLP}(\mathbf{q}^{t-1} \parallel \mathbf{p}^{t-1}))$

$\mathbf{p}^t \leftarrow (\mathbf{1} - \mathbf{g}^t) \odot \mathbf{p}^{t-1} + \mathbf{g}^t \odot \mathbf{f}^t$

$\mathbf{q}^t \leftarrow \mathbf{q}^{t-1} + \mathbf{p}^t$

end for

$\hat{\mathbf{X}}_r \leftarrow \mathbf{X}_r + \alpha_{\text{aux}} \mathbf{q}^T$

$\hat{\mathbf{X}}_d \leftarrow \mathbf{X}_d + \alpha_{\text{main}} \mathbf{q}^T$

Asymmetric residual fusion. After T refinement steps, we obtain $\mathbf{q}^T$ as a refined auxiliary state. We then output

$$\hat{\mathbf{X}}_r = \mathbf{X}_r + \alpha_{\text{aux}} \mathbf{q}^T, \quad (12)$$

$$\hat{\mathbf{X}}_d = \mathbf{X}_d + \alpha_{\text{main}} \mathbf{q}^T. \quad (13)$$

where $\alpha_{\text{aux}}, \alpha_{\text{main}} \in \mathbb{R}$ are learnable scalars. Setting $\alpha_{\text{main}} = 0$ enforces strict block-level one-way fusion (no auxiliary-to-primary update), while a nonzero α_{main} permits auxiliary-to-primary injection whose sensitivity is quantified in Corollary 3.6.

Algorithm. Algorithm 1 summarizes one HAF block.

3. Finite-Horizon Guarantees

The HAF block is unrolled for only a small number of steps ($T = 4$ in the reported experiments). Its relevant robustness property is therefore finite-horizon sensitivity rather than asymptotic convergence to a stationary state. We compare the exact rollout driven by a clean auxiliary input with the rollout driven by its corrupted counterpart, while keeping the primary input fixed. The analysis preserves the complete state-dependent gate and accounts for auxiliary corruption entering through both the initial query and the auxiliary key used at every refinement step.

Importantly, the results below are finite-horizon and conditional. They do not assert unconditional asymptotic stability, energy conservation, or global contraction of the learned dynamics.

3.1. Clean-corrupted trajectory formulation

Let ξ denote the clean auxiliary input and let $\xi + \varepsilon$ denote its corrupted counterpart. The primary input and its key $\mathbf{K}_d$ are held fixed. The auxiliary input determines both the initial query $\mathbf{q}^0(\xi) = \mathbf{Q}_r(\xi)$ and the auxiliary key $\mathbf{K}_r(\xi)$.

For fixed keys, SGC is real-linear in its query argument. Define

$$\mathcal{A}_{\mathbf{K}}(\mathbf{q}) := \text{SGC}(\mathbf{q}, \mathbf{K}).$$

The exact force operator associated with the auxiliary input ξ is

$$\mathbf{M}(\xi) := \mathcal{A}_{\mathbf{K}_d} + (\gamma - 1)\mathbf{I} - \gamma\mathcal{A}_{\mathbf{K}_r(\xi)}. \quad (14)$$

Consequently, the implemented force can be written exactly as

$$\mathbf{f}(\mathbf{q}; \xi) = \mathbf{M}(\xi)\mathbf{q}. \quad (15)$$

We identify feature tensors with their vectorizations whenever Jacobians are written. For the product state $\mathbf{z} = (\mathbf{q}, \mathbf{p})$, define

$$\mathbf{D}(\mathbf{z}) := \text{Diag}(\text{vec } \mathbf{g}(\mathbf{z})). \quad (16)$$

The momentum update and the complete state update can then be represented by

$$\begin{aligned} \psi(\mathbf{z}; \xi) &:= (\mathbf{I} - \mathbf{D}(\mathbf{z}))\mathbf{p} + \mathbf{D}(\mathbf{z})\mathbf{M}(\xi)\mathbf{q}, \\ \Phi(\mathbf{z}; \xi) &:= (\mathbf{q} + \psi(\mathbf{z}; \xi), \psi(\mathbf{z}; \xi)). \end{aligned} \quad (17)$$

Thus, the exact rollout satisfies

$$\mathbf{z}^t(\xi) = \Phi(\mathbf{z}^{t-1}(\xi); \xi). \quad (18)$$

The subscript t allows the local derivative bounds to vary across unroll steps, even when the block parameters are shared.

On the product state space, we use the norm

$$\|\mathbf{z}\|_{\mathcal{Z}} := (\|\mathbf{q}\|_F^2 + \|\mathbf{p}\|_F^2)^{1/2}. \quad (19)$$

The complete state Jacobian can be written explicitly. Define

$$\mathbf{r}(\mathbf{z}; \xi) := \mathbf{M}(\xi)\mathbf{q} - \mathbf{p}. \quad (20)$$

Since

$$\psi = \mathbf{p} + \mathbf{g}(\mathbf{z}) \odot \mathbf{r}(\mathbf{z}; \xi),$$

let

$$\begin{aligned} \mathbf{B}_q(\mathbf{z}; \xi) &:= \mathbf{D}(\mathbf{z})\mathbf{M}(\xi) + \text{Diag}(\mathbf{r}(\mathbf{z}; \xi))D_{\mathbf{q}}\mathbf{g}(\mathbf{z}), \\ \mathbf{B}_p(\mathbf{z}; \xi) &:= \mathbf{I} - \mathbf{D}(\mathbf{z}) + \text{Diag}(\mathbf{r}(\mathbf{z}; \xi))D_{\mathbf{p}}\mathbf{g}(\mathbf{z}). \end{aligned} \quad (21)$$

Writing

$$\mathbf{J}(\mathbf{z}; \xi) := D_{\mathbf{z}}\Phi(\mathbf{z}; \xi),$$

we obtain

$$\mathbf{J}(\mathbf{z}; \xi) = \begin{bmatrix} \mathbf{I} + \mathbf{B}_q & \mathbf{B}_p \\ \mathbf{B}_q & \mathbf{B}_p \end{bmatrix}. \quad (22)$$

This Jacobian retains both the position–momentum coupling and every derivative of the learned state-dependent gate.

3.2. Finite-horizon sensitivity bound

Define the clean and corrupted trajectories as

$$\mathbf{z}_c^t := \mathbf{z}^t(\xi), \quad \mathbf{z}_\varepsilon^t := \mathbf{z}^t(\xi + \varepsilon). \quad (23)$$

Assumption 3.1 (Rollout regularity). For each $t = 1, \dots, T$, there exists a convex set $\mathcal{R}_t \subset \mathcal{Z} \times \mathcal{X}$ containing the following two line segments:

$$\Gamma_t^{\text{state}} := \{((1-s)\mathbf{z}_c^{t-1} + s\mathbf{z}_\varepsilon^{t-1}, \xi + \varepsilon) : s \in [0, 1]\}, \quad (24)$$

$$\Gamma_t^{\text{input}} := \{(\mathbf{z}_c^{t-1}, \xi + s\varepsilon) : s \in [0, 1]\}. \quad (25)$$

Assume that Φ_t is continuously differentiable on $\mathcal{R}_t$, and define

$$\begin{aligned} a_t &:= \sup_{(\mathbf{z}, \xi) \in \mathcal{R}_t} \|D_{\mathbf{z}}\Phi_t(\mathbf{z}; \xi)\|_{\mathcal{Z} \rightarrow \mathcal{Z}} < \infty, \\ b_t &:= \sup_{(\mathbf{z}, \xi) \in \mathcal{R}_t} \|D_{\xi}\Phi_t(\mathbf{z}; \xi)\|_{\mathcal{X} \rightarrow \mathcal{Z}} < \infty. \end{aligned} \quad (26)$$

For locally Lipschitz network components that are piecewise continuously differentiable, as in standard finite neural networks, the same assumption and conclusions hold with the derivative suprema interpreted as essential suprema along the relevant segments, or equivalently with the corresponding segmentwise Lipschitz factors.

Moreover, the initial state satisfies

$$\|\mathbf{z}^0(\xi + \varepsilon) - \mathbf{z}^0(\xi)\|_{\mathcal{Z}} \leq c_0\|\varepsilon\|_{\mathcal{X}}. \quad (27)$$

All bounds below are uniform over the selected sets $\mathcal{R}_t$. If these sets are constructed only for one clean–corrupted trajectory pair, the resulting certificate is trajectory-wise. A global certificate requires fixed sets covering the entire admissible corruption class.

Theorem 3.2 (Finite-horizon sensitivity bound). *Under Assumption 3.1, the final refined query satisfies*

$$\|\mathbf{q}_\varepsilon^T - \mathbf{q}_c^T\|_F \leq C_T\|\varepsilon\|_{\mathcal{X}}, \quad (28)$$

where, with the empty product defined as one,

$$C_T := c_0 \prod_{j=1}^T a_j + \sum_{\ell=1}^T b_\ell \prod_{j=\ell+1}^T a_j. \quad (29)$$

The two terms in (29) represent two distinct corruption paths. The first term propagates corruption entering through the initial query. The second term accumulates corruption repeatedly injected through the auxiliary key at different refinement steps. Theorem 3.2 therefore accounts for all auxiliary-input paths in the implemented rollout, including their indirect effects through the state-dependent gate.

Proposition 3.3 (HAF-specific direct-input bound). *Suppose that the auxiliary-key projection $\mathbf{K}_r$ is L_K -Lipschitz. Assume also that*

$$\|\mathbf{D}(\mathbf{z})\|_{2 \rightarrow 2} \leq g_{\max, t}$$

on $\mathcal{R}_t$. For a query state $\mathbf{q}$, define

$$C_{\mathbf{q}} := \|\mathbf{G}\|_{\infty} \|\mathbf{w}\|_{\infty} \|\mathcal{F}(\mathbf{q} \odot \mathbf{w})\|_{\infty}, \quad (30)$$

and let

$$\overline{C}_{q, t} := \sup_{(\mathbf{z}, \xi) \in \mathcal{R}_t} C_{\mathbf{q}(\mathbf{z})} < \infty, \quad (31)$$

where $\mathbf{q}(\mathbf{z})$ denotes the query component of $\mathbf{z}$. Then the direct auxiliary-input factor in (26) satisfies

$$b_t \leq \sqrt{2} \gamma g_{\max, t} \overline{C}_{q, t} L_K. \quad (32)$$

The bound follows from the key-argument regularity of SGC and is specific to the repeated auxiliary-key corruption path in HAF. The factor $\sqrt{2}$ appears because the same directly induced momentum perturbation enters both components of the product state $(\mathbf{q}, \mathbf{p})$.

3.3. Error attenuation

Corollary 3.4 (Uniform finite-horizon bound). *If*

$$a_t \leq a < 1, \quad b_t \leq b$$

for every t , then

$$C_T \leq c_0 a^T + b \frac{1 - a^T}{1 - a}. \quad (33)$$

The first term is the attenuated contribution of the initial-query perturbation. The second term is the finite accumulated contribution caused by repeated injection through the corrupted auxiliary key.

The requirement $a < 1$ is an additional sufficient condition rather than an unconditional property of the HAF architecture. In particular, it is not implied solely by the sigmoid gate or the leaky momentum update and must be verified or explicitly enforced for the trained rollout. Even when $a < 1$ holds, it does not imply contraction of the complete rollout perturbation because the direct-input factors b_t may remain nonzero. A precise contraction statement therefore requires comparison with the actual initial-state perturbation.

Corollary 3.5 (Sufficient condition for refined-query perturbation contraction). *Suppose, in addition to Assumption 3.1, that a considered nonzero corruption direction satisfies*

$$\Delta_{\varepsilon} \mathbf{z}^0 := \mathbf{z}^0(\xi + \varepsilon) - \mathbf{z}^0(\xi) \quad (34)$$

and

$$\|\Delta_{\varepsilon} \mathbf{z}^0\|_{\mathcal{Z}} \geq c_0 \|\varepsilon\|_{\mathcal{X}} \quad (35)$$

for some direction-dependent constant $c_0 > 0$. If

$$C_T < c_0, \quad (36)$$

then

$$\begin{aligned} \|\mathbf{q}_{\varepsilon}^T - \mathbf{q}_c^T\|_F &\leq \eta_T \|\Delta_{\varepsilon} \mathbf{z}^0\|_{\mathcal{Z}}, \\ \eta_T &:= \frac{C_T}{c_0} < 1. \end{aligned} \quad (37)$$

Theorem 3.2 provides a finite-horizon sensitivity bound. By contrast, Corollary 3.5 asserts refined-query contraction only for a considered corruption direction satisfying both (35) and (36). The lower-gain condition excludes directions that are annihilated by the initial query projection.

Furthermore, because the auxiliary output in (12) retains a direct residual connection from the auxiliary input, contraction of the refined query does not by itself imply contraction of the complete auxiliary output. More precisely, when $\xi = \mathbf{X}_r$ and the same Frobenius norm is used,

$$\hat{\mathbf{X}}_{r, \varepsilon} - \hat{\mathbf{X}}_{r, c} = \varepsilon + \alpha_{\text{aux}} (\mathbf{q}_{\varepsilon}^T - \mathbf{q}_c^T). \quad (38)$$

Therefore,

$$\|\hat{\mathbf{X}}_{r, \varepsilon} - \hat{\mathbf{X}}_{r, c}\|_F \leq (1 + |\alpha_{\text{aux}}| C_T) \|\varepsilon\|_{\mathcal{X}}. \quad (39)$$

Thus, the contraction statement concerns the refined query state rather than the complete residual auxiliary output.

3.4. Primary-stream safety

Corollary 3.6 (Block-level primary perturbation bound). *Let the primary input be identical in the clean and auxiliary-corrupted forward passes. Under Theorem 3.2, define*

$$\begin{aligned} \Delta_{\varepsilon} \mathbf{q}^T &:= \mathbf{q}_{\varepsilon}^T - \mathbf{q}_c^T, \\ \Delta_{\varepsilon} \hat{\mathbf{X}}_d &:= \hat{\mathbf{X}}_{d, \varepsilon} - \hat{\mathbf{X}}_{d, c}. \end{aligned} \quad (40)$$

Then

$$\begin{aligned} \|\Delta_{\varepsilon} \hat{\mathbf{X}}_d\|_F &= |\alpha_{\text{main}}| \|\Delta_{\varepsilon} \mathbf{q}^T\|_F \\ &\leq |\alpha_{\text{main}}| C_T \|\varepsilon\|_{\mathcal{X}}. \end{aligned} \quad (41)$$

If $\alpha_{\text{main}} = 0$, then the primary output is exactly independent of the auxiliary input. This constitutes strict block-level one-way safety. For nonzero α_{main} , (41) provides an explicit finite-horizon perturbation bound, but it does not imply strict one-way independence.

Corollary 3.6 is a single-block statement. For a network containing multiple HAF blocks, a stack-level sensitivity bound must be obtained by composing the sensitivity factors of the intervening blocks and transformations. Exact stack-level one-way independence is preserved when every auxiliary-to-primary residual path is structurally disabled.

Table 1. Quantitative comparison on six RGB-D saliency datasets. Metrics include S-Measure (S_m), max F-Measure (F_β), max E-Measure (E_ξ), and MAE (M). $\uparrow/\downarrow$ indicates higher/lower is better. Best results are in **bold**, and second-best results are underlined. Gray highlights **our method (HAF)**.

Method	DUT-RGBD				LFSD				NJU2K				NLPR				SIP				STERE			
	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
SwinNet (Liu et al., 2021)	0.945	0.941	0.967	0.023	0.878	0.868	0.899	0.060	0.931	0.919	0.929	0.028	0.941	0.908	0.967	0.018	<u>0.911</u>	0.912	0.943	0.035	-	-	-	-
MutualFormer (Wang et al., 2024)	0.936	0.946	0.966	0.024	0.872	0.879	0.911	0.062	0.922	0.923	0.954	0.032	0.932	0.925	0.965	0.021	0.894	0.902	0.932	0.043	0.908	0.908	0.947	0.038
CIRNet (Cong et al., 2022)	0.932	0.937	0.952	0.028	0.875	0.882	0.902	0.067	0.925	0.927	0.925	0.035	0.934	0.924	0.955	0.027	0.888	0.895	0.917	0.052	0.916	0.913	0.928	0.037
CPNet (Hu et al., 2024)	<u>0.949</u>	<u>0.960</u>	<u>0.975</u>	<u>0.019</u>	0.881	0.881	<u>0.916</u>	<u>0.057</u>	0.936	0.943	<u>0.966</u>	<u>0.024</u>	0.936	0.931	0.969	0.019	0.906	<u>0.926</u>	<u>0.945</u>	0.035	<u>0.923</u>	0.921	<u>0.958</u>	0.029
CATNet (Sun et al., 2023)	0.953	0.951	0.971	0.020	0.894	<u>0.884</u>	0.908	0.051	<u>0.937</u>	0.929	0.933	0.025	0.939	0.916	0.968	0.018	0.913	0.918	0.944	<u>0.034</u>	0.925	0.902	0.935	0.030
HFMDNet (Luo et al., 2024)	-	-	-	-	0.880	0.883	0.915	0.059	<u>0.937</u>	<u>0.944</u>	<u>0.966</u>	0.023	0.938	0.933	<u>0.971</u>	<u>0.017</u>	0.886	0.905	0.930	0.044	0.918	0.920	0.957	0.031
EM-Trans (Chen et al., 2025)	-	-	-	-	-	-	-	-	0.931	0.935	0.961	0.027	<u>0.940</u>	<u>0.934</u>	0.970	<u>0.017</u>	0.903	0.920	0.944	0.039	0.925	<u>0.926</u>	<u>0.958</u>	<u>0.028</u>
HAF (Ours)	0.953	0.963	0.979	0.017	<u>0.888</u>	0.890	0.923	0.051	0.938	0.945	0.967	0.023	<u>0.940</u>	0.938	0.974	0.016	0.913	0.933	0.951	0.033	0.925	0.928	0.962	0.027

Table 2. Ablation on fusion strategies. Comparison of HAF against alternative fusion variants on six datasets using S_m , max F_β , max E_ξ , and MAE (M). $\uparrow/\downarrow$ indicates higher/lower is better. Best results are in **bold**, and second-best results are underlined. Gray highlights **our method (HAF)**.

Method	DUT-RGBD				LFSD				NJU2K				NLPR				SIP				STERE			
	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
+LMF-Net (Li et al., 2025)	0.945	0.956	0.972	0.021	<u>0.889</u>	<u>0.887</u>	<u>0.923</u>	<u>0.051</u>	0.931	0.938	<u>0.964</u>	0.026	0.940	<u>0.936</u>	0.974	0.016	0.908	0.925	<u>0.949</u>	<u>0.034</u>	0.926	0.923	0.958	<u>0.028</u>
+EM-Trans (Chen et al., 2025)	<u>0.948</u>	<u>0.959</u>	<u>0.974</u>	<u>0.020</u>	0.891	0.890	0.925	0.050	<u>0.935</u>	0.940	<u>0.964</u>	<u>0.024</u>	0.937	0.933	0.971	0.018	<u>0.912</u>	<u>0.929</u>	<u>0.949</u>	0.033	0.922	<u>0.924</u>	<u>0.960</u>	0.029
+MA (Yu et al., 2025)	0.946	0.953	0.970	0.024	0.877	0.877	0.913	0.061	0.928	0.936	0.960	0.028	0.932	0.927	0.968	0.018	0.894	0.915	0.940	0.041	0.916	0.916	0.954	0.033
+CA (Yu et al., 2025)	0.946	0.957	0.973	<u>0.020</u>	0.879	0.880	0.916	0.055	0.933	0.938	0.962	0.025	<u>0.939</u>	0.935	<u>0.973</u>	<u>0.017</u>	0.901	0.920	0.940	0.040	0.917	0.919	0.957	0.031
+SHIP (Zhou et al., 2025)	0.947	0.958	0.973	<u>0.020</u>	0.883	0.881	0.921	0.055	<u>0.935</u>	<u>0.942</u>	<u>0.964</u>	0.025	0.936	0.931	0.969	0.020	0.907	0.927	0.948	<u>0.034</u>	0.921	0.921	0.957	0.030
HAF (Ours)	0.953	0.963	0.979	0.017	0.888	0.890	<u>0.923</u>	<u>0.051</u>	0.938	0.945	0.967	0.023	0.940	0.938	0.974	0.016	0.913	0.933	0.951	0.033	<u>0.925</u>	0.928	0.962	0.027

4. Experiments

Experiment design around modality imbalance and directed refinement. Experiments are organized to evaluate whether *Hamiltonian Asymmetric Fusion (HAF)* improves robustness under *modality imbalance*. In this regime, symmetric bidirectional fusion may permit *noise backflow* from a noisier stream into a designated primary representation, potentially exacerbating degradation under multi-step updates. Experiments therefore assess whether role-asymmetric refinement with explicit primary-influence control mitigates such deterioration. Evidence in the main paper is structured as follows: (i) overall comparisons against recent RGB-D SOD methods on six benchmarks using standard metrics; (ii) a controlled module-swapping protocol that replaces *only* the fusion operator while keeping the backbone, training protocol, and hyperparameters unchanged; (iii) refinement dynamics analyzed by varying the unroll step number T and visualizing step-wise cross-modal interaction maps; and (iv) robustness to auxiliary-modality corruption summarized by *Degradation AUC* over increasing corruption levels (lower is better) and shown as cross-dataset box plots. Additional results are deferred to Appendix B.

4.1. Experimental Setup

Datasets. We evaluate on six standard RGB-D salient object detection benchmarks: DUT-RGBD (Piao et al., 2019), LFSD (Li et al., 2014), NJU2K (Wang et al., 2015), NLPR (Peng et al., 2014), SIP (Fan et al., 2020) and STERE (Niu et al., 2012). These datasets span diverse capture conditions (e.g., consumer light-field cameras, structured-light depth sensors, and stereo-derived depth) and include challenging

cases such as small salient objects, complex backgrounds, and person-centric scenes. For the fair comparison, we take the same training dataset as in (Hu et al., 2024), including 2,985 images (1,485 from NJU2K, 700 from NLPR, and 800 from DUT-RGBD) and report results on all six test sets.

Implementation details. Unless otherwise stated, we follow CPNet (Hu et al., 2024) architecture. We replace its original fusion blocks at multiple hierarchical stages with HAF, while leaving all other components unchanged, to model cross-modal correspondence and progressively refine the fused representations. We set $\gamma = 0.4$ in all experiments. A lightweight sensitivity check that varies γ under a single-step update with all other settings fixed indicates that $\gamma = 0.4$ is a robust default across datasets. We train with Adam using a batch size of 16 and an initial learning rate of 5×10^{-5} , decayed by a factor of 10 every 100 epochs. Training converges within 150 epochs on a single NVIDIA RTX A6000 GPU. Code is available at [MaITL-Group/HAF](https://github.com/MaITL-Group/HAF).

Evaluation metrics. We follow the standard RGB-D SOD protocol and report four commonly used metrics: S_m (Fan et al., 2017), E_ξ (Fan et al., 2018), max F_β (Achanta et al., 2009), and MAE (Perazzi et al., 2012). S_m and E_ξ evaluate structural and alignment-aware similarity, max F_β summarizes the precision-recall trade-off, and MAE measures the pixel-wise absolute error. For each benchmark, we use its official evaluation script with the default settings to ensure consistent and comparable reporting.

Baselines. (1) We compare against recent RGB-D SOD methods that represent dominant design patterns for transformer-based multimodal fusion and progressive decoding, including SwinNet (Liu et al., 2021), MutualFormer

Table 3. **Role-swap ablation on four datasets.** “Ours” uses the default primary/auxiliary assignment in HAF (Depth as primary, RGB as auxiliary). “Swap” reverses the primary/auxiliary roles (RGB as primary, Depth as auxiliary), while keeping all other settings and the evaluation protocol identical to the main setting.

Model	DUT-RGBD				NJU2K				NLPR				SIP			
	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
Ours	0.953	0.963	0.979	0.017	0.938	0.945	0.967	0.023	0.940	0.938	0.974	0.016	0.913	0.933	0.951	0.033
Swap	0.951	0.961	0.977	0.017	0.935	0.940	0.964	0.025	0.938	0.935	0.972	0.017	0.905	0.927	0.945	0.035

P/A denote primary/auxiliary. “Swap” only reverses the P/A roles (default: D→P, R→A; swapped: R→P, D→A).

(Wang et al., 2024), CIRNet (Cong et al., 2022), CPNet (Hu et al., 2024), CATNet (Sun et al., 2023), HFMDNet (Luo et al., 2024) and EM-Trans (Chen et al., 2025). Except for CPNet, whose results are reproduced by running the released source code, all other results are taken from the corresponding official reported numbers. (2) Different fusion methods, including LMF-Net (Li et al., 2025), EM-Trans (Chen et al., 2025), MA (Merge-Attention) (Yu et al., 2025), CA (Co-Attention) (Yu et al., 2025), and SHIP (Zhou et al., 2025), are evaluated by replacing the fusion module in CPNet while keeping all other settings unchanged.

4.2. Comparisons With State-of-The-Arts

Quantitative Results. Tables 1 report results on six RGB-D SOD benchmarks using S_m , max F_β , max E_ξ , and MAE. HAF performs competitively across datasets and metrics, often ranking among the top methods. Gains on boundary-sensitive measures (F_β , E_ξ) together with lower MAE are consistent with improved delineation and reduced spurious responses. These trends are observed across benchmarks with diverse object scales and background complexity.

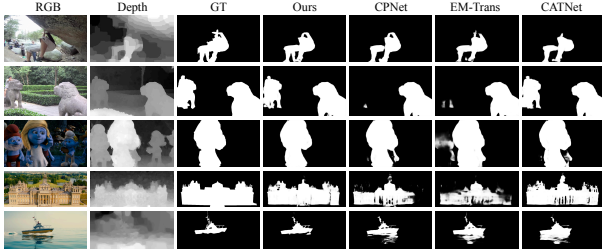


Figure 3. **Qualitative comparison** between our HAF and recent state-of-the-art RGB-D SOD methods.

Qualitative Results. Fig. 3 shows comparisons on diverse scenes (e.g., cluttered backgrounds, thin structures, and low-contrast foregrounds). Across the shown examples, HAF exhibits more localized foreground activation and reduced background leakage. Compared with baselines shown in the figure (e.g., CPNet and EM-Trans), predictions tend to show reduced fragmentation while preserving fine details, including thin structures and low-contrast regions.

4.3. Ablation Studies

1) Ablation on Fusion Strategies. To isolate the fusion design, the fusion module is replaced with representative alternatives (LMF-Net, EM-Trans, MA, CA, and SHIP) while keeping the backbone, training protocol, and evaluation unchanged. Table 2 summarizes results on all six benchmarks. Under this controlled module-swapping setting, HAF shows consistent improvements across datasets and metrics, suggesting that gains primarily stem from the fusion mechanism rather than unrelated architectural or training differences.

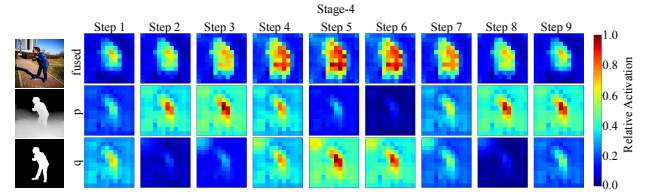


Figure 4. **Effect of iteration steps.** Step-wise interaction maps at Stage-4 for different unroll steps T . As steps increase, activations tend to concentrate on salient regions with reduced background responses; the most effective T varies across datasets, and visualizations for all stages are provided in the appendix.

2) Effect of Iteration Steps. The effect of the unroll step number T is examined. Fig. 4 shows the step-wise evolution of interaction maps at Stage-4; visualizations for other stages are deferred to the Appendix B.5. As T increases, activations tend to become more spatially coherent and concentrate on salient regions, while background responses are reduced. Gains typically saturate beyond a moderate number of steps, and very large T yields limited additional benefit. Throughout the experiments, T is fixed at 4.

3) Role-swap ablation. Table 3 evaluates the sensitivity of HAF to the task-defined primary/auxiliary (P/A) assignment. “Ours” follows the main setting (Depth as primary, RGB as auxiliary), while “Swap” reverses the P/A roles with all other settings and evaluation unchanged. Swapping roles generally reduces performance on the evaluated benchmarks. Overall, the depth-primary assignment is preferable under the current RGB-D SOD protocol, while the swapped variant remains competitive.

4) SGC vs. MHSA Alternative. To validate the choice of spectral global correlation (SGC) in our refinement operator,

Table 4. SGC vs. MHSA ablation. Comparison of SGC and MHSA in efficiency and performance.

Method	Params(M)	FLOPs(G)	NJU2K				SIP			
			$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
SGC	232.328	130.1	0.938	0.945	0.967	0.023	0.913	0.933	0.951	0.033
MHSA	232.365	208.7	0.933	0.939	0.962	0.025	0.911	0.931	0.950	0.034

we replace SGC with a multi-head self-attention (MHSA) based implementation while keeping the backbone, training schedule, and evaluation protocol unchanged. As shown in Table 4, SGC consistently yields better performance on both datasets (e.g., higher $S_m/F_\beta/E_\xi$ and lower MAE), indicating that the spectral correlation operator is a more effective global interaction primitive in our setting. Meanwhile, SGC is also more efficient, requiring fewer FLOPs than the MHSA alternative at comparable parameter count, supporting its use as a practical drop-in component.

5) Efficiency. Table 5 reports the efficiency comparison. CPNet+CDA is constructed by replacing only the original CPNet fusion module with CDA, while keeping all other components unchanged. This replacement causes only a moderate increase in parameters but results in extremely large FLOPs, indicating that dense cross-diffusion attention is computationally prohibitive when applied to multi-scale high-resolution features. In comparison, HAF keeps the computation close to the original CPNet baseline, increasing FLOPs only from 129.3G to 130.1G, while using fewer parameters than CPNet+CDA. These results demonstrate that HAF achieves a more favorable efficiency trade-off for directed multimodal fusion.

Table 5. Comparison of model efficiency. ‘‘CPNet+CDA’’ denotes a controlled variant where the original feature fusion module in CPNet is replaced by the CDA module, while other components are kept unchanged. FLOPs are reported per image.

Method	Params (M)	FLOPs (G)
CIRNet (Cong et al., 2022)	103.2	42.6
CATNet (Sun et al., 2023)	262.6	341.8
HFMDNet (Luo et al., 2024)	431.6	242.2
CPNet (Hu et al., 2024)	216.5	129.3
CPNet+CDA (Hu et al., 2024; Wang et al., 2024)	244.9	26099.7
HAF(Ours)	232.3	130.1

6) Robustness to Auxiliary-Modality Corruption. To summarize noise sensitivity beyond per-level curves, robustness is quantified by *Degradation AUC* of F_β (lower is better). The score is computed on the discrete corruption schedule as the area under the non-negative performance-drop curve relative to the clean setting. Fig. 5 reports the cross-dataset distribution for different fusion operators. HAF attains the lowest median AUC with a tight interquartile range, while several baselines exhibit larger AUC or higher dispersion, consistent with greater sensitivity or stronger dataset dependence under the same schedule.

Additional ablations. Additional analyses of frequency response, gate behavior, efficiency, transferability, robustness,

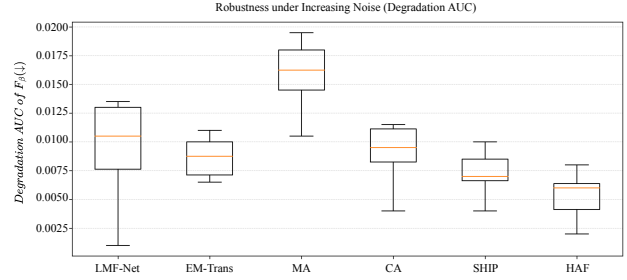


Figure 5. Robustness under increasing corruption (Degradation AUC). Degradation AUC integrates the non-negative drop in F_β from the clean setting over the discrete noise schedule (lower is better). Boxplots show the cross-dataset distribution; HAF achieves the lowest median AUC with low dispersion.

and qualitative comparisons are provided in Appendix B.

5. Conclusion

This paper studies multimodal fusion under modality imbalance, where symmetric interactions may allow auxiliary noise to affect the primary stream during iterative refinement. *Hamiltonian Asymmetric Fusion* (HAF) performs role-asymmetric directed refinement, updating auxiliary tokens under primary guidance with explicitly quantified auxiliary-to-primary influence. An accompanying analysis derives a finite-horizon refined-query sensitivity bound, a direction-dependent sufficient condition under which the refined-query perturbation contracts relative to the induced initial-state perturbation, and a block-level primary perturbation bound; exact primary independence holds when $\alpha_{\text{main}} = 0$. Experiments on six RGB-D SOD benchmarks show improved accuracy and more graceful degradation under controlled auxiliary-modality corruption.

6. Limitations

Evaluation focuses on RGB-D SOD with fixed modality roles and a fixed refinement step; preliminary RGB-T transfer results are provided in the supplement, while broader multimodal evaluation remains future work. The method assumes task-relevant reliability asymmetry, requiring role validation when this asymmetry is weak or ambiguous.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Appendix

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A. Related Work

A.1. RGB-D Salient Object Detection

RGB-D salient object detection (SOD) has progressed from CNN-based two-stream encoders with lightweight, hand-crafted fusion patterns to Transformer and hybrid backbones that emphasize long-range context modeling and modality-aware interaction. Early RGB-D SOD methods focused on cross-modal fusion to exploit depth geometry while mitigating noisy measurements, spanning cascaded, parallel, recurrent, and attention-based designs; examples include SwinNet (Liu et al., 2021), CCAFNet (Zhou et al., 2021b), C2DFNet (Zhang et al., 2023), and selective/self-mutual attention variants (Wang et al., 2022).

In the Transformer era, a common pipeline adopts dual-stream encoders to extract modality-specific tokens, followed by an explicit fusion module and progressive decoding for stage-wise refinement. CPNet is representative in this line, formulating RGB–depth interaction through cross-modal fusion and refining saliency maps progressively (Hu et al., 2024). CATNet further emphasizes cascaded aggregation and correction to alleviate information loss and scale mismatch in hierarchical Transformer features (Sun et al., 2023), while HFMDNet focuses on hierarchical fusion with multilevel decoding to better exploit complementarity across feature hierarchies (Luo et al., 2024). Beyond fusion/decoding design, EM-Trans strengthens boundary localization via edge-aware learning, reducing contour ambiguity for dense prediction (Chen et al., 2025).

In a closely related direction, MutualFormer proposes Cross-Diffusion Attention (CDA), which reframes cross-modal interaction by constructing cross-affinities from modality-specific affinities in a metric space as an alternative to standard cross-attention (Wang et al., 2024). While CDA improves the interaction *operator*, its symmetric cross-diffusion treats both modalities on equal footing and is role-agnostic with respect to primary–auxiliary roles, which may be less suited for role-asymmetric settings where modality reliability differs (e.g., noisy or missing depth). This motivates fusion mechanisms that can model cross-modal interaction while explicitly accounting for primary–auxiliary roles, rather than enforcing fully symmetric coupling.

A.2. Multimodal Fusion Modules

Multimodal fusion can be categorized by where interaction occurs (early, intermediate, late) and whether the goal is explicit alignment or task-driven complementarity extraction (Li & Tang, 2026). In modern deep models, intermediate (feature-level) fusion is widely used because it preserves modality-specific encoders while enabling learnable cross-modal interaction, making it well-suited for plug-and-play fusion modules in diverse multimodal settings.

Within feature-level fusion, practical designs often start from additive or concatenation-based aggregation and then introduce adaptive operators such as attention or gating to suppress noisy modalities and emphasize complementary cues. LMF-Net (Li et al., 2025) combines multiscale pooling with learnable embedding fusion and feature fusion to dynamically adjust cross-modal collaboration in dense prediction. More expressive interaction paradigms have recently moved beyond second-order relationships: SHIP models higher-order cross-modal interactions across spatial and channel dimensions, motivated by the limited expressiveness of standard cross-attention (Zhou et al., 2025). Meanwhile, evidence suggests that multimodal information is distributed across feature hierarchies, and fusion outcomes can be sensitive to both fusion position and fusion pattern (Lin et al., 2025), motivating progressive and iterative fusion modules in practice.

Despite these advances, many fusion modules primarily enhance interaction expressiveness (e.g., attention, gating, higher-order modeling), while explicitly modeling modality roles and reliability remains less explored. These plug-and-play designs enable controlled *module-swapping* under a unified backbone/decoder setting, making it easier to analyze how fusion blocks trade off interaction expressiveness against robustness under modality degradation.

A.3. Physics-inspired Neural Models

A natural perspective for iterative refinement is to treat representation update as a dynamical process, where features evolve through structured update rules rather than ad hoc multi-stage operations. Physics-guided and dynamical-inspired learning injects inductive biases (e.g., invariance, stability, conservation) into neural models, often improving robustness and interpretability, while raising challenges in optimization and scalable deployment (Wang & Yu, 2025).

Hamiltonian formulations provide a representative mechanism for structure-preserving evolution. Hamiltonian Neural Networks (HNNs) parameterize a Hamiltonian function and derive dynamics via Hamilton’s equations, yielding improved long-term behavior due to built-in conservation properties (Greydanus et al., 2019). To reduce numerical drift, symplectic

learning frameworks incorporate geometric priors and symplectic integration, e.g., Symplectic Recurrent Neural Networks (SRNNs) for stable simulation of Hamiltonian systems (Chen et al., 2020). In vision and representation learning, continuous-depth models such as Neural ODEs interpret feature transformations as solutions of differential equations, providing a principled view of iterative refinement; related ideas have also been explored in generative modeling (Toth et al., 2020).

Different from prior work that primarily targets simulation, forecasting, or generation, these models suggest a dynamics-inspired route for designing iterative update rules (Wang & Yu, 2025). Building on this perspective, it is natural to explore unrolled, dynamics-inspired fusion modules that can regulate multi-step interactions for cross-modal refinement, potentially in a role-aware (primary–auxiliary) manner. HAF adopts this design motivation but uses a gated leaky-momentum update rather than claiming exact Hamiltonian conservation or symplecticity.

B. Additional Experimental Results

B.1. Ablation on the learnable frequency response $\mathbf{G}$.

Table 6 evaluates the impact of the learnable frequency response $\mathbf{G}$ in SGC. The full model (w/ $\mathbf{G}$) is compared with a variant that disables spectral reweighting by setting $\mathbf{G} \equiv \mathbf{1}$ (w/o $\mathbf{G}$), with all other components and the training/evaluation protocol unchanged. Removing $\mathbf{G}$ generally degrades performance across the evaluated datasets, with lower S_m , F_β , and E_ξ and higher MAE. These results suggest that learnable spectral reweighting improves the flexibility of the global correlation operator and provides more informative cues for refinement.

Table 6. Ablation on the learnable frequency filter $\mathbf{G}$ in SGC. “w/ $\mathbf{G}$ ” uses the learnable spectral modulation in Eq. (1), while “w/o $\mathbf{G}$ ” disables it by setting $\mathbf{G} \equiv \mathbf{1}$ (i.e., no learnable frequency reweighting). All other components, training settings, and the evaluation protocol are kept identical to the main setting.

Model	DUT-RGBD				NJU2K				NLPR				SIP			
	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
w/o $\mathbf{G}$	0.950	0.961	0.976	0.019	0.935	0.941	0.963	0.025	0.936	0.933	0.970	0.018	0.908	0.932	0.949	0.034
w/ $\mathbf{G}$	0.953	0.963	0.979	0.017	0.938	0.945	0.967	0.023	0.940	0.938	0.974	0.016	0.913	0.933	0.951	0.033

B.2. Gate Input Ablation.

Gate inputs are ablated on four datasets by feeding only the state $\mathbf{q}$, only the momentum $\mathbf{p}$, or their concatenation $[\mathbf{q}; \mathbf{p}]$, while keeping all other settings identical to the main configuration (Table 7). On four evaluated datasets, $[\mathbf{q}; \mathbf{p}]$ yields the best overall results, whereas using a single term generally degrades performance across metrics. This indicates that conditioning the gate on both the current state ($\mathbf{q}$) and the momentum term ($\mathbf{p}$), which reflects recent update dynamics, provides richer context for modulating auxiliary-driven updates in multi-step refinement.

Table 7. Ablation of the gate input in HAF on four representative datasets. We compare using only the state $\mathbf{q}$, only the momentum $\mathbf{p}$, and their concatenation $[\mathbf{q}; \mathbf{p}]$ as the gate input, while keeping all other settings identical to the main setting.

Gate input	DUT-RGBD				NJU2K				NLPR				SIP			
	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$M \downarrow$
$\mathbf{q}$	0.951	0.962	0.977	0.019	0.935	0.942	0.964	0.024	0.937	0.932	0.969	0.019	0.908	0.929	0.947	0.034
$\mathbf{p}$	0.952	0.963	0.977	0.018	0.937	0.943	0.966	0.023	0.938	0.935	0.970	0.019	0.913	0.933	0.951	0.032
$[\mathbf{q}; \mathbf{p}]$	0.953	0.963	0.979	0.017	0.938	0.945	0.967	0.023	0.940	0.938	0.974	0.016	0.913	0.933	0.951	0.033

B.3. Gate distribution diagnostics.

To assess whether the learned gate exhibits obvious saturation (i.e., mass concentrated near 0/1), Table 8 reports simple distribution statistics of the Stage-4 gate outputs. Following the implementation, per-image gate mean μ_g and standard deviation σ_g are computed over all spatial positions and channels, and then aggregated as $\mathbb{E}[\mu_g]$, $\text{Std}(\mu_g)$, and $\mathbb{E}[\sigma_g]$. We also report the saturation rate $\text{Sat.}(\%)$, defined as the fraction of gate values within τ of the extremes (e.g., $g < \tau$ or $g > 1 - \tau$, using $\tau = (0.05/0.95)$). Across both datasets, Sat. remains near zero while $\mathbb{E}[\sigma_g]$ indicates non-trivial variability, which is consistent with non-saturated, input-dependent gating under the main setting.

Table 8. Empirical diagnostics of the gate values at stage 4 for the adopted gate design. We summarize the distribution of per-image gate statistics over the dataset.

Dataset	$\mathbb{E}[\mu_g]$	$\text{Std}(\mu_g)$	$\mathbb{E}[\sigma_g]$	Sat.(%) $\downarrow$
NJU2K	0.481	0.002	0.034	0.00
SIP	0.482	0.001	0.033	0.00

B.4. Compute Cost vs. Unroll Steps.

We profile the compute cost of HAF as a function of the unroll steps T and report the estimated per-image FLOPs in Table 9. Over the evaluated range ($T \in [1, 9]$), the cost scales approximately linearly with T ; increasing T from 1 to 9 adds only 2.07 G FLOPs per image at an input resolution of 384^2 with $B=1$, implying an approximately constant per-step increment under the same FLOPs estimator. This near-linear behavior provides a straightforward way to budget per-image compute when selecting T in the reported setting.

Table 9. Full FLOPs vs. unroll steps T ($size = 384^2$, $B = 1$).

T	1	2	3	4	5	6	7	8	9
FLOPs (G)	129.286	129.545	129.804	130.062	130.321	130.580	130.839	131.097	131.356

B.5. Stage-wise Interaction Evolution

Fig. 6 complements the main visualization by reporting step-wise interaction maps for all stages (Stage-1~Stage-4). In the shown examples, earlier stages (Stage-1/2) emphasize fine-grained edge/texture patterns, while deeper stages (Stage-3/4) exhibit increasingly object-centric activations, aligning with the hierarchical representations of the backbone. Across refinement steps, background responses are often attenuated and salient regions become more spatially coherent, although the saturation point can vary across stages and datasets. Overall, these stage-wise trends provide qualitative context for adopting a moderate (and dataset-dependent) unroll depth in the main evaluations.

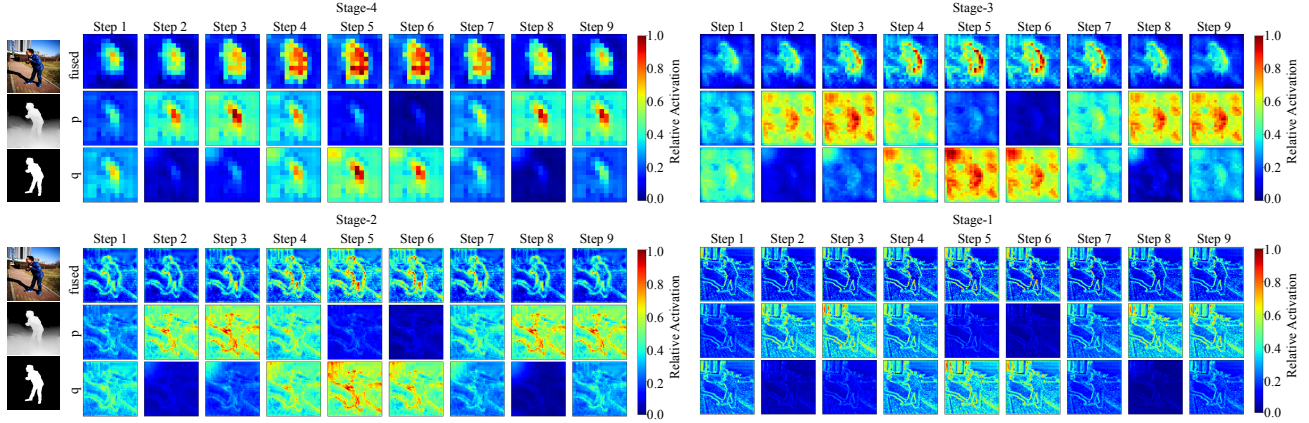


Figure 6. **Step-wise evolution across network stages.** Visualization of the interaction maps over refinement steps ($t=1 \dots 9$) at all four stages (Stage-1~Stage-4). For each stage, rows correspond to the fused map and the modality-specific interaction components (p and q), with color indicating relative activation. Across steps, the interaction responses tend to become more spatially structured and concentrate on salient regions, while the most effective refinement depth can vary by stage and dataset.

B.6. Extension to RGB-T SOD.

To further examine whether the proposed directed-refinement principle can transfer beyond RGB-D SOD, we additionally evaluate HAF as a plug-in fusion replacement in an RGB-T saliency pipeline based on ContriNet (Tang et al., 2025). Following the same reliability-guided role assignment rule, we use the thermal modality as the primary stream and RGB as the auxiliary stream. As shown in Table 10, HAF consistently improves ContriNet on VT821 and VT1000 across all reported metrics, and also brings small but consistent gains on VT5000. These results provide preliminary evidence that HAF is not limited to RGB-D fusion, but can also serve as an effective directed-refinement module in another bimodal saliency setting. We note that broader multimodal transfer remains future work.

Table 10. Transfer ablation on RGB-T saliency detection. HAF is evaluated as a plug-in fusion replacement in the ContriNet pipeline, where the thermal modality is used as the primary stream and RGB is used as the auxiliary stream following the same reliability-guided principle. We report S_m , F_β , F_β^w , E_ξ , and $\mathcal{M}$, where higher is better for all metrics except $\mathcal{M}$.

Method	VT821					VT1000					VT5000				
	$S_m \uparrow$	$F_\beta \uparrow$	$F_\beta^w \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$F_\beta^w \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	$F_\beta^w \uparrow$	$E_\xi \uparrow$	$\mathcal{M} \downarrow$
ContriNet (Tang et al., 2025)	0.879	0.836	0.819	0.912	0.033	0.925	0.903	0.901	0.949	0.019	0.894	0.864	0.848	0.934	0.029
HAF (Ours)	0.888	0.843	0.833	0.922	0.029	0.929	0.905	0.906	0.950	0.018	0.895	0.864	0.849	0.936	0.028

B.7. Robustness under Increasing Corruption Levels

To complement the cross-dataset summary in the main paper, additional per-dataset visualizations are provided here. Fig. 7 plots the level-wise change in F_β across the corruption schedule, highlighting how performance varies as noise increases. Fig. 8 summarizes the same trend with a single robustness score via degradation AUC, computed from the non-negative drop relative to the clean setting. Under this protocol, HAF tends to show smaller accumulated degradation (lower AUC) and milder declines in ΔF_β on the shown datasets, consistent with improved robustness to auxiliary corruption.

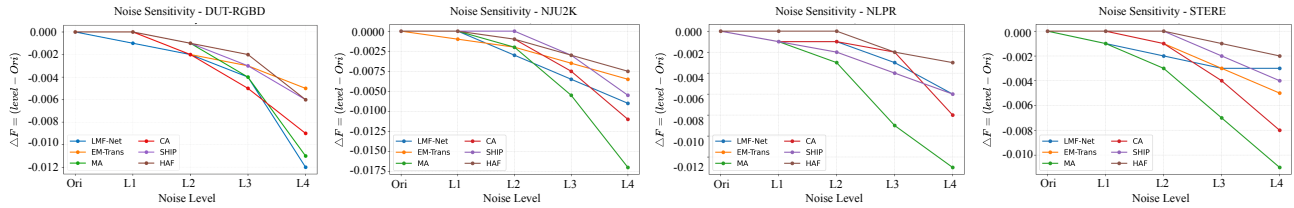


Figure 7. Level-wise degradation under increasing auxiliary corruption. Performance change under auxiliary corruption, reported as $\Delta F_\beta(l) = F_\beta(l) - F_\beta(\text{Ori})$ over noise levels $l \in \{L1-L4\}$. Values closer to 0 indicate smaller deviation from the clean setting; negative values correspond to improvements relative to Ori.

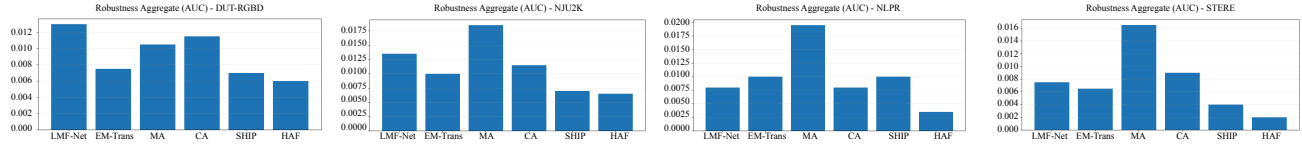


Figure 8. Aggregated robustness via degradation AUC. Bars report degradation AUC, computed on the discrete noise levels (trapezoidal rule) from the non-negative drop $\max(0, F_\beta(\text{Ori}) - F_\beta(l))$. Lower AUC indicates smaller accumulated degradation as corruption increases.

B.8. Additional Qualitative Comparisons

Figure 9 provides additional qualitative comparisons on RGB-D SOD. Across a range of scenes, the predictions from HAF typically preserve spatially coherent foreground regions and well-localized object extents, while exhibiting reduced background responses and fewer fragmented regions. The examples cover challenging conditions such as cluttered backgrounds, thin structures, and low-contrast foregrounds, where alternative methods may show partial missing regions, over-smoothed boundaries, or scattered activations. These visualizations serve as complementary evidence to the main-paper results under diverse appearance and depth configurations.

C. Proofs

This appendix provides the complete derivations supporting Section 3. Feature tensors are identified with their vectorizations whenever Jacobians or induced operator norms are used. Accordingly, the Frobenius norm of a feature tensor coincides with the Euclidean norm of its vectorization, and the product-state norm is defined by (19). All linearity statements below are understood over the real feature space used by the model.

C.1. SGC regularity in the query and key arguments

Lemma C.1 (Boundedness and joint regularity of SGC). *Assume unitary FFT normalization and the real-valued SGC convention in (1). For fixed $\mathbf{K}$, define*

$$\mathcal{A}_{\mathbf{K}}(\mathbf{Q}) := \text{SGC}(\mathbf{Q}, \mathbf{K}).$$

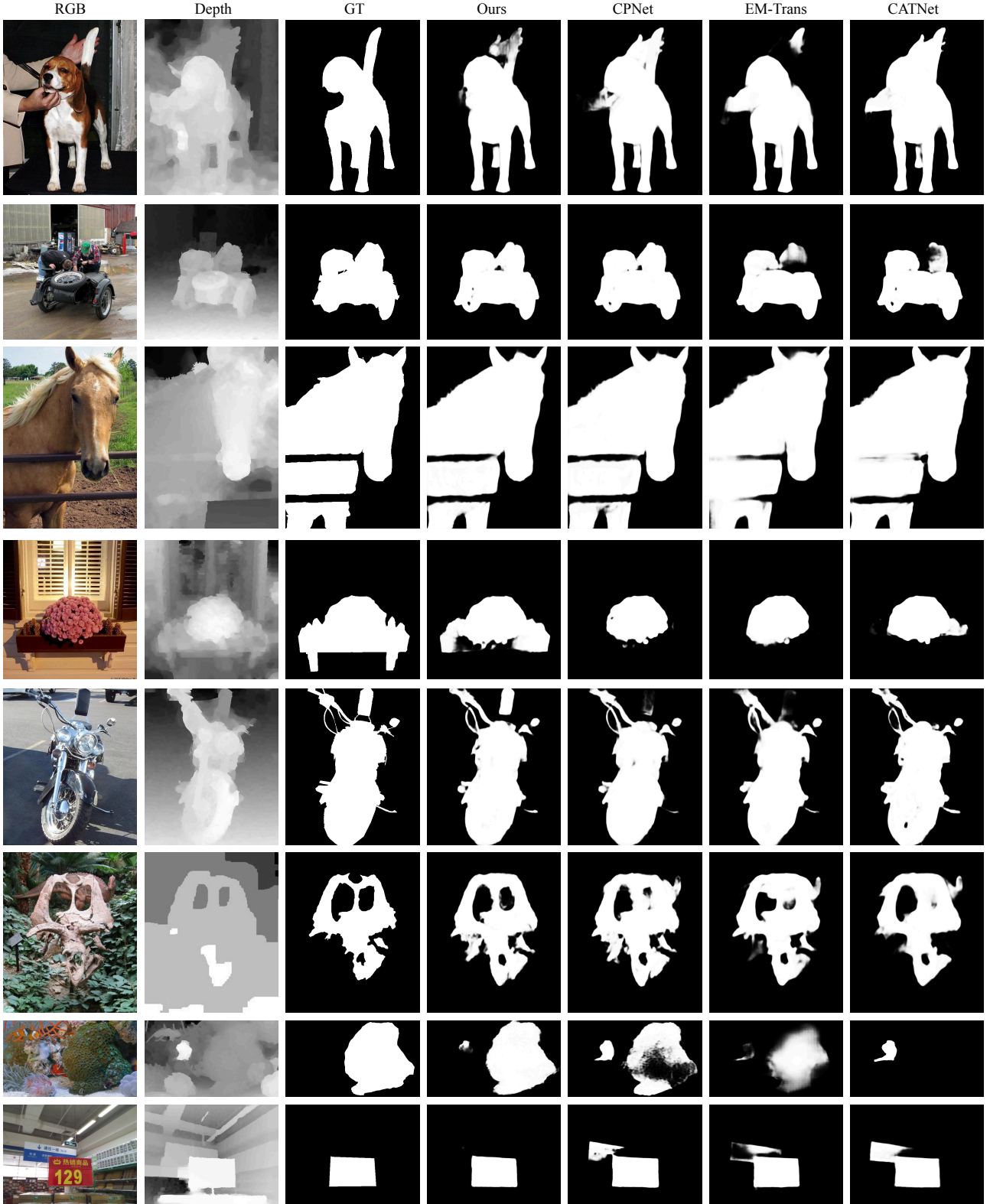


Figure 9. Qualitative comparison between our HAF and recent state-of-the-art RGB-D SOD methods.

Then $\mathcal{A}_{\mathbf{K}}$ is real-linear and satisfies

$$\|\mathcal{A}_{\mathbf{K}}\|_{2 \rightarrow 2} \leq \|\mathbf{G}\|_{\infty} \|\mathbf{w}\|_{\infty} \|\mathcal{F}(\mathbf{K} \odot \mathbf{w})\|_{\infty}. \quad (42)$$

Moreover, for fixed $\mathbf{Q}$,

$$\begin{aligned} \|\text{SGC}(\mathbf{Q}, \mathbf{K}_1) - \text{SGC}(\mathbf{Q}, \mathbf{K}_2)\|_F &\leq C_{\mathbf{Q}} \|\mathbf{K}_1 - \mathbf{K}_2\|_F, \\ C_{\mathbf{Q}} &:= \|\mathbf{G}\|_{\infty} \|\mathbf{w}\|_{\infty} \|\mathcal{F}(\mathbf{Q} \odot \mathbf{w})\|_{\infty}. \end{aligned} \quad (43)$$

Consequently, SGC is jointly Lipschitz on bounded query–key sets.

Proof. Let $\Delta\mathbf{Q} = \mathbf{Q}_1 - \mathbf{Q}_2$. Real-linearity follows directly from the real-valued SGC definition. Using the non-expansiveness of taking the real part, Parseval’s identity under unitary FFT normalization, and the standard product inequality gives

$$\begin{aligned} \|\mathcal{A}_{\mathbf{K}}(\Delta\mathbf{Q})\|_F &\leq \|\mathbf{G} \odot \mathcal{F}(\Delta\mathbf{Q} \odot \mathbf{w}) \odot \overline{\mathcal{F}(\mathbf{K} \odot \mathbf{w})}\|_F \\ &\leq \|\mathbf{G}\|_{\infty} \|\mathcal{F}(\mathbf{K} \odot \mathbf{w})\|_{\infty} \|\mathcal{F}(\Delta\mathbf{Q} \odot \mathbf{w})\|_F \\ &\leq \|\mathbf{G}\|_{\infty} \|\mathbf{w}\|_{\infty} \|\mathcal{F}(\mathbf{K} \odot \mathbf{w})\|_{\infty} \|\Delta\mathbf{Q}\|_F. \end{aligned} \quad (44)$$

This proves (42).

For fixed $\mathbf{Q}$, let $\Delta\mathbf{K} = \mathbf{K}_1 - \mathbf{K}_2$. The same calculation yields

$$\begin{aligned} \|\text{SGC}(\mathbf{Q}, \mathbf{K}_1) - \text{SGC}(\mathbf{Q}, \mathbf{K}_2)\|_F &\leq \|\mathbf{G}\|_{\infty} \|\mathcal{F}(\mathbf{Q} \odot \mathbf{w})\|_{\infty} \|\mathcal{F}(\Delta\mathbf{K} \odot \mathbf{w})\|_F \\ &\leq \|\mathbf{G}\|_{\infty} \|\mathbf{w}\|_{\infty} \|\mathcal{F}(\mathbf{Q} \odot \mathbf{w})\|_{\infty} \|\Delta\mathbf{K}\|_F, \end{aligned} \quad (45)$$

which proves (43).

Finally, split

$$\text{SGC}(\mathbf{Q}_1, \mathbf{K}_1) - \text{SGC}(\mathbf{Q}_2, \mathbf{K}_2) = \text{SGC}(\mathbf{Q}_1, \mathbf{K}_1) - \text{SGC}(\mathbf{Q}_2, \mathbf{K}_1) + \text{SGC}(\mathbf{Q}_2, \mathbf{K}_1) - \text{SGC}(\mathbf{Q}_2, \mathbf{K}_2).$$

On bounded query–key sets, the two one-argument Lipschitz factors are uniformly bounded. Applying the triangle inequality and the preceding bounds therefore proves joint Lipschitzness. $\square$

C.2. Derivation of the exact HAF rollout Jacobian

From (17),

$$\psi(\mathbf{z}; \xi) = \mathbf{p} + \mathbf{D}(\mathbf{z})(\mathbf{M}(\xi)\mathbf{q} - \mathbf{p}).$$

Let

$$\mathbf{r}(\mathbf{z}; \xi) = \mathbf{M}(\xi)\mathbf{q} - \mathbf{p}.$$

For a state perturbation $d\mathbf{z} = (d\mathbf{q}, d\mathbf{p})$, the product rule gives

$$d\psi = \mathbf{D}(\mathbf{z})\mathbf{M}(\xi) d\mathbf{q} + (\mathbf{I} - \mathbf{D}(\mathbf{z})) d\mathbf{p} + \text{Diag}(\mathbf{r}) (D_{\mathbf{q}}\mathbf{g}(\mathbf{z}) d\mathbf{q} + D_{\mathbf{p}}\mathbf{g}(\mathbf{z}) d\mathbf{p}).$$

Therefore,

$$D_{\mathbf{q}}\psi = \mathbf{B}_q, \quad D_{\mathbf{p}}\psi = \mathbf{B}_p,$$

where $\mathbf{B}_q$ and $\mathbf{B}_p$ are defined in (21). Since

$$\Phi = (\mathbf{q} + \psi, \psi),$$

stacking the two derivatives yields (22). This derivation retains both the position–momentum coupling and all derivatives of the state-dependent gate.

When the state $\mathbf{z}$ is held fixed, the gate has no direct dependence on ξ . Hence

$$D_{\xi}\psi = \mathbf{D}(\mathbf{z})[D_{\xi}\mathbf{M}(\xi)]\mathbf{q}, \quad D_{\xi}\Phi = \begin{bmatrix} D_{\xi}\psi \\ D_{\xi}\psi \end{bmatrix}. \quad (46)$$

C.3. Proof of Theorem 3.2

Proof. Let

$$\mathbf{e}_t := \mathbf{z}_\varepsilon^t - \mathbf{z}_c^t.$$

Adding and subtracting $\Phi(\mathbf{z}_c^{t-1}; \xi + \varepsilon)$ gives

$$\mathbf{e}_t = \Phi(\mathbf{z}_\varepsilon^{t-1}; \xi + \varepsilon) - \Phi(\mathbf{z}_c^{t-1}; \xi + \varepsilon) + \Phi(\mathbf{z}_c^{t-1}; \xi + \varepsilon) - \Phi(\mathbf{z}_c^{t-1}; \xi).$$

For the first difference, apply the mean-value inequality along the state segment

$$s \mapsto ((1-s)\mathbf{z}_c^{t-1} + s\mathbf{z}_\varepsilon^{t-1}, \xi + \varepsilon), \quad s \in [0, 1].$$

For the second difference, apply it along the input segment

$$s \mapsto (\mathbf{z}_c^{t-1}, \xi + s\varepsilon), \quad s \in [0, 1].$$

Both segments are contained in $\mathcal{R}_t$ by Assumption 3.1. Therefore,

$$\|\mathbf{e}_t\|_{\mathcal{Z}} \leq a_t \|\mathbf{e}_{t-1}\|_{\mathcal{Z}} + b_t \|\varepsilon\|_{\mathcal{X}}. \quad (47)$$

Using (27) and repeatedly substituting (47), we obtain

$$\begin{aligned} \|\mathbf{e}_T\|_{\mathcal{Z}} &\leq \left(c_0 \prod_{j=1}^T a_j + \sum_{\ell=1}^T b_\ell \prod_{j=\ell+1}^T a_j \right) \|\varepsilon\|_{\mathcal{X}} \\ &= C_T \|\varepsilon\|_{\mathcal{X}}. \end{aligned} \quad (48)$$

Projection onto the $\mathbf{q}$ component has operator norm one under (19). Consequently,

$$\|\mathbf{q}_\varepsilon^T - \mathbf{q}_c^T\|_F \leq \|\mathbf{e}_T\|_{\mathcal{Z}},$$

which proves (28). $\square$

C.4. Proof of Proposition 3.3

Proof. Fix a query state $\mathbf{q}$. The primary-key term and the identity term in $\mathbf{M}(\xi)$ are unchanged under auxiliary corruption. Therefore, for $\xi' = \xi + \varepsilon$,

$$\begin{aligned} \|\mathbf{M}(\xi') - \mathbf{M}(\xi)\|_F &= \gamma \|\text{SGC}(\mathbf{q}, \mathbf{K}_r(\xi')) - \text{SGC}(\mathbf{q}, \mathbf{K}_r(\xi))\|_F \\ &\leq \gamma C_{\mathbf{q}} \|\mathbf{K}_r(\xi') - \mathbf{K}_r(\xi)\|_F \\ &\leq \gamma C_{\mathbf{q}} L_K \|\varepsilon\|_{\mathcal{X}}, \end{aligned} \quad (49)$$

where the first inequality follows from (43).

At fixed $\mathbf{z}$, only ψ changes directly with the auxiliary input. From the exact state map,

$$\Phi(\mathbf{z}; \xi') - \Phi(\mathbf{z}; \xi) = \begin{bmatrix} \delta_\psi \\ \delta_\psi \end{bmatrix}, \quad \delta_\psi := \mathbf{D}(\mathbf{z})[\mathbf{M}(\xi') - \mathbf{M}(\xi)]\mathbf{q}.$$

Consequently,

$$\begin{aligned} \|\Phi(\mathbf{z}; \xi') - \Phi(\mathbf{z}; \xi)\|_{\mathcal{Z}} &= \sqrt{2} \|\mathbf{D}(\mathbf{z})[\mathbf{M}(\xi') - \mathbf{M}(\xi)]\mathbf{q}\|_F \\ &\leq \sqrt{2} \gamma g_{\max, t} C_{\mathbf{q}} L_K \|\varepsilon\|_{\mathcal{X}}. \end{aligned} \quad (50)$$

At every differentiability point, divide (50) by $\|\varepsilon\|_{\mathcal{X}}$ and take the directional limit as $\varepsilon \rightarrow \mathbf{0}$. This yields

$$\|D_{\xi} \Phi(\mathbf{z}; \xi)\|_{\mathcal{X} \rightarrow \mathcal{Z}} \leq \sqrt{2} \gamma g_{\max, t} C_{\mathbf{q}} L_K.$$

Taking the supremum over $(\mathbf{z}, \xi) \in \mathcal{R}_t$ gives

$$b_t \leq \sqrt{2} \gamma g_{\max, t} \bar{C}_{q, t} L_K,$$

which is (32). For piecewise continuously differentiable network components, the same argument applies to the essential supremum, or equivalently to the corresponding segmentwise Lipschitz factor. $\square$

C.5. Proof of Corollaries 3.4 and 3.5

Proof. If $a_t \leq a < 1$ and $b_t \leq b$ for every t , then

$$c_0 \prod_{j=1}^T a_j \leq c_0 a^T$$

and

$$\sum_{\ell=1}^T b_\ell \prod_{j=\ell+1}^T a_j \leq b \sum_{k=0}^{T-1} a^k = b \frac{1-a^T}{1-a}.$$

Combining the two inequalities proves (33).

For the contraction statement, (35) implies

$$\|\varepsilon\|_{\mathcal{X}} \leq \frac{1}{\underline{c}_0} \|\Delta_\varepsilon \mathbf{z}^0\|_{\mathcal{Z}}.$$

Combining this inequality with (28) gives

$$\|\mathbf{q}_\varepsilon^T - \mathbf{q}_c^T\|_F \leq \frac{C_T}{\underline{c}_0} \|\Delta_\varepsilon \mathbf{z}^0\|_{\mathcal{Z}}.$$

Thus

$$\eta_T = \frac{C_T}{\underline{c}_0},$$

and the strict inequality $\eta_T < 1$ follows exactly from $C_T < \underline{c}_0$. This proves (37). $\square$

C.6. Proof of Corollary 3.6

Proof. The primary input is identical in the clean and auxiliary-corrupted forward passes, and

$$\hat{\mathbf{X}}_d = \mathbf{X}_d + \alpha_{\text{main}} \mathbf{q}^T.$$

Consequently,

$$\Delta_\varepsilon \hat{\mathbf{X}}_d = \alpha_{\text{main}} (\mathbf{q}_\varepsilon^T - \mathbf{q}_c^T).$$

Taking Frobenius norms and applying Theorem 3.2 gives

$$\|\Delta_\varepsilon \hat{\mathbf{X}}_d\|_F = |\alpha_{\text{main}}| \|\mathbf{q}_\varepsilon^T - \mathbf{q}_c^T\|_F \leq |\alpha_{\text{main}}| C_T \|\varepsilon\|_{\mathcal{X}},$$

which proves (41).

If $\alpha_{\text{main}} = 0$, then

$$\Delta_\varepsilon \hat{\mathbf{X}}_d = \mathbf{0}$$

for every admissible auxiliary perturbation. Hence the primary output is exactly independent of the auxiliary input at the block level. $\square$